History of Integrated Circuits video presentation slides (2)

Condensed Study Notes

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LESSON ONE: INTEGRATED CIRCUIT (IC) Video Presentation

The development of Integrated Circuits (ICs) was driven by two main needs:

1. **Software Advancement:** Rapid growth in software sophistication required matching hardware innovation. Without it, software progress would be limited by the processing power of existing electronics.

2. **Hardware Limitations:** Early mainframe computers were extremely large due to their reliance on vacuum tubes. These systems also consumed a significant amount of electricity.

COMPUTER HARDWARE

Computer hardware refers to the physical components of a computer system. These are the tangible parts you can see and touch.

Early computers relied on vacuum tubes for their electronic functions. These tubes were large, consumed significant power, and generated a lot of heat. The development of Integrated Circuits (ICs) was a crucial advancement. ICs miniaturized electronic components onto a single chip, leading to smaller, more efficient, and more powerful computers.

Key components of computer hardware include:

- **Central Processing Unit (CPU):** The 'brain' of the computer, responsible for executing instructions.
- **Memory (RAM):** Temporary storage for data and programs actively being used by the CPU.
- **Storage Devices:** Permanent storage for data and programs (e.g., hard drives, SSDs).
- **Input/Output (I/O) Devices:** Allow interaction with the computer (e.g., keyboard, mouse, monitor).

3.0 Introduction to Integrated Circuits (ICs

Integrated Circuits (ICs), also known as chips, microchips, computer circuits, microprocessor chips, or silicon chips, are assemblies of microscopic electronic components like transistors, diodes, capacitors, and resistors. These components and their interconnections are fabricated as a single unit on a semiconducting material, typically silicon.

The primary component within an IC is the transistor. Early ICs in the late 1950s contained around 10 components on a tiny chip. The development of Very Large-Scale Integration (VLSI) dramatically increased the number of components that could be placed on a chip, leading to the creation of the microprocessor.

This miniaturization allows for hundreds, thousands, millions, or even billions of components to be integrated onto a chip no larger than a fingernail. The first commercially successful IC chip (Intel, 1974) had 4,800 transistors, a number that grew to 3.2 million in Intel's Pentium (1993) and now exceeds a billion.

2.0 History of Integrated Circuits

The Integrated Circuit (IC) was invented independently by Jack Kilby and Robert Noyce.

- Jack Kilby worked for Texas Instruments (USA).
- Robert Noyce worked for Fairchild Semiconductor.

Kilby filed his patent first, but Noyce's was granted first, leading to a legal dispute. Their companies eventually agreed to share the technology, with a joint patent granted in 1960.

Both Kilby and Noyce were recognized for their groundbreaking invention, being inducted into the inventor's Hall of Fame.

The IC is considered a highly significant and far-reaching technology of the 20th century.

3.1 Fitting IC to a Circuit board

Integrated Circuits (ICs) are physically placed onto circuit boards. A bread circuit board is shown with an IC inserted.

Key identification for IC placement:

- Pin 1 of an IC is marked by a notch.

Power connections:

- The black wire is connected to GND (Ground).
- The red wire is connected to Vcc (Voltage supply pin).

5.0 Classification of Integration Circuit

Integrated Circuits (ICs) are classified based on two main criteria:

1. **Type of Transistors Employed:**
    - **Bipolar Integrated Circuits:** Use bipolar junction transistors. Generally chosen for the highest logic speed.
    - **MOS (Metal-Oxide-Semiconductor) Integrated Circuits:** Generally used for large-scale integration or lowest power dissipation. High-performance bipolar transistors and complementary MOS (CMOS) transistors can be combined (BiCMOS) for high speed and density.
    - Both types rely on creating patterns of electrically active impurities within the semiconductor and forming metal interconnections on its surface.
2. **Scale of Integration (Number of Transistors/Active Devices):**
    - **Small-Scale Integration (SSI):** Fewer than 10 transistors.
    - **Medium-Scale Integration (MSI):** 10 to 100 transistors.

- **Large-Scale Integration (LSI):** 100 to 1,000 transistors.
- **Very-Large-Scale Integration (VLSI):** More than 1,000 transistors.

4.0 Types of IC

ICs are primarily categorized by the nature of the signals they process: Analog and Digital.

1. **Analog ICs** (also called Linear ICs):
    - Handle signals that are continuous in nature, meaning they can take on any value within a range.
    - The output signal's level is directly dependent on the input signal's level.
    - Examples include Operational Amplifiers (Op Amps), Voltage Regulators, and Comparators.
2. **Digital ICs**:
    - Process signals in a discrete, binary format, representing data as either a logical '0' or a logical '1'.
    - Typically, '0' corresponds to 0 Volts and '1' corresponds to +5 Volts.
    - Examples include Logic Gates, Counters, and Flip-flops.

6.0 Usage of IC

Integrated Circuits (ICs) are versatile components that can perform a wide range of functions within electronic systems.

Key functions of ICs include:

- Amplifier: Increases the strength of a signal.
- Oscillator: Generates repetitive electronic signals.
- Timer: Controls the timing of events.
- Counter: Counts events or pulses.
- Memory: Stores digital information.
- Microprocessor: The central processing unit (CPU) of a computer, capable of executing instructions.

7.0 Discrete Vs Integrated Circuit

Discrete circuits are built by connecting individual electronic components like Bipolar Junction Transistors (BJTs), Field Effect Transistors (FETs), diodes, resistors, and capacitors.

Each of these components is a separate physical part that must be wired together to form a functional circuit.

8.0 What is Doping Semiconductors

Doping involves adding small quantities of impurities to a silicon atom. This process creates two types of semiconductors:

- **N-type:** Doping silicon with phosphorus results in an N-type semiconductor where electrons predominate.
- **P-type:** Doping silicon with boron results in a P-type semiconductor where holes predominate.

8.1 P–N Junction

A p–n junction is the interface within a single semiconductor crystal where p-type and n-type materials meet.

- **p-type material**: Has an excess of holes (positively charged vacancies).
- **n-type material**: Has an excess of electrons (negatively charged particles).

Both materials start as electrically neutral atoms.

8.2 P-N Junction Diode

A P-N junction diode is created by fusing together N-type and P-type semiconductor materials.

In a circuit, the function of a diode is to...

9.0 IC Fabrication Process

IC fabrication begins with silicon, a raw element that is chemically treated or 'doped' to alter its electrical properties.

Doping introduces impurities to silicon, creating N-type (electron-rich) and P-type (hole-rich) semiconductor materials.

The fabrication process requires extremely clean environments called 'clean rooms' to prevent contamination, as even tiny specks of dirt can cause defects at the microscopic scale. Workers in clean rooms wear protective clothing and pass through airlocks.

Doping can be achieved through various methods:

- Sputtering: Ions of the doping material are propelled at the silicon wafer.
- Vapor Deposition: The doping material is introduced as a gas, which condenses to form a thin film of impurities on the wafer.
- Molecular Beam Epitaxy: A highly precise deposition technique.

a) The processes involved in IC fabrication

IC fabrication involves six main steps, some of which are repeated.

1. **Wafer Making**: Pure silicon crystals are grown into cylinders, then sliced thinly like salami into wafers. Each wafer will be cut into many individual chips.

2. **Masking**: Wafers are heated and coated with silicon dioxide. Ultraviolet light is used to apply a protective layer called photoresist.

3. **Etching**: A chemical removes parts of the photoresist, creating a template pattern that defines where specific semiconductor types (N-type or P-type) will be formed.

4. **Doping**: Etched wafers are heated with impurity-containing gases to create N-type (electron-rich) and P-type (hole-rich) silicon areas. This step may be followed by more masking and etching.

5. **Testing**: Computer-controlled testers connect to each chip's terminals. Non-functional chips are marked and rejected.

6. **Packaging**: Working chips are cut from the wafer and encased in protective plastic, preparing them for integration into electronic devices.

9.1 A summary of IC Fabrication Process

IC fabrication is a highly simplified, multi-step process. It involves creating the physical silicon wafer, precisely patterning it, doping it to form semiconductor properties, testing the resulting chips, and finally packaging them.

Key stages include:

- Wafer creation
- Patterning (masking and etching)
- Doping
- Testing
- Packaging

b) Quality Control

Despite rigorous controls and precise tools in IC fabrication, a significant number of chips are inevitably rejected.

While the percentage of rejected chips has decreased over time, the complexity of creating microscopic interwoven circuits means some level of failure remains unavoidable.

c) Hazardous Materials and Recycling

The fabrication of ICs involves the use of dopants like gallium and arsenic, which are toxic. Strict protocols are necessary for their storage, use, and disposal.

Due to their widespread applicability, a significant recycling industry exists for ICs. Many integrated circuits and other electronic components are salvaged from obsolete equipment, tested, and then resold for use in new devices.